The University of Texas at Austin
School of Information

Week Five

Techniques and Frameworks

Prompt Engineering

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Week Five

Agenda

- Presentations: Vanshita, Jaykumar
- News
- Review whatialreadyknow (Ishwari)
- eB review
- eC preview
- m1 questions
- Agenta (again)
- Potato (again)
- Techniques
- Frameworks
- Work time

Presentations

News

s1

- From `simplescaling`
- Trained on 1,000 examples
- Each example is ⟨ a question ⟩ ⟨ an answer ⟩ ⟨ a reasoning process ⟩
- First tried 59,000 examples
- article at Economist

The Batch

⟨ pause to look at this week's edition ⟩

WhatIAlreadyKnow (Ishwari)

eB review

Smart strategies

- adjust the temperature (needs API)
- adjust the max_tokens (needs API)
- do it piecemeal
- use more than one LLM
- think about the goal more than constraints

eC preview

⟨ look at the doc ⟩

m1 questions

Deliverable

- A short qmd / html document describing the domain
- The doc should include specification of whether you plan to run locally or using a cloud service
- The doc should include a discussion of the possible datasets you might use (the actual dataset will be due in m2)
- You are not required to stick with the directions you give here, but this should be your current best guess of what you plan to do
- Examples: chatbot to emulate a foreign leader; chatbot to triage banking problems; chatbot to analyze tweets

My message

⟨ look at the message ⟩

Agenta (again)

Potato (again)

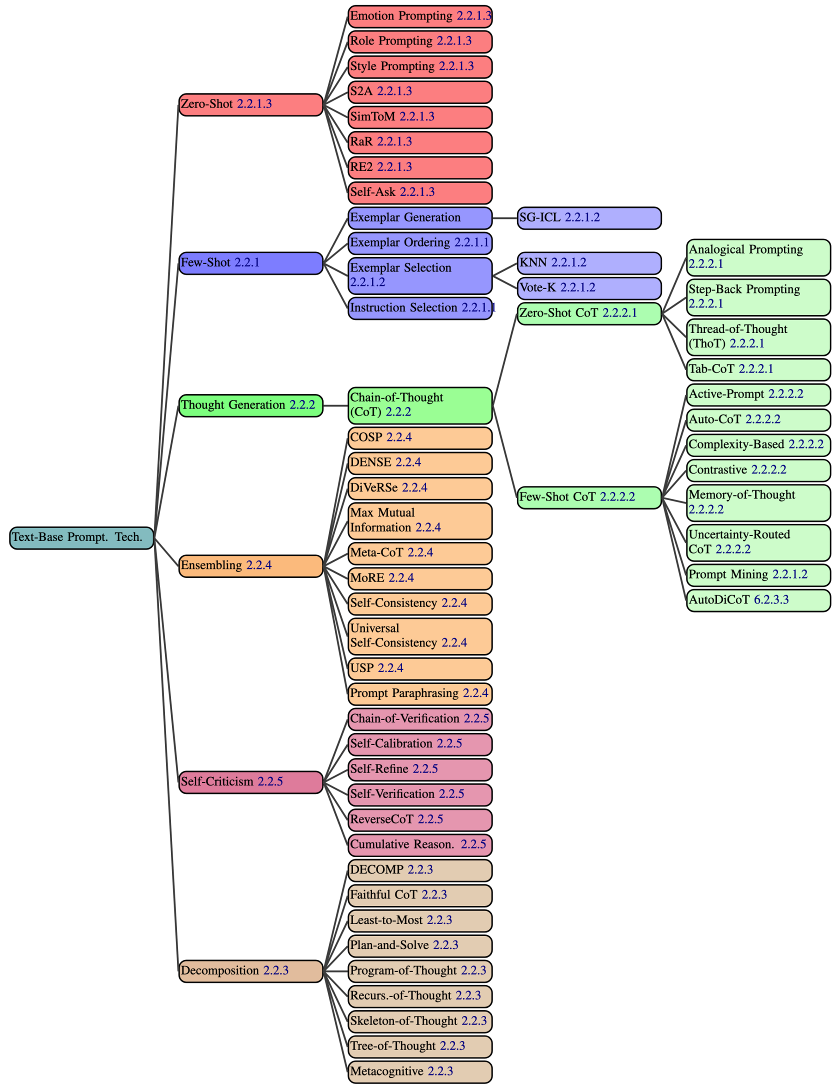
```
mkdir labeling && cd labeling
git clone https://github.com/davidjurgens/potato.git
cd potato
pip install -r requirements.txt
python potato/flask_server.py start project-hub/simple-examples/configs/simple-check-box.yaml -
```

This will allow you to login. Then you can go to `http://localhost:8000` and login. Then try the following while you're still logged in:

```
python potato/flask_server.py start project-hub/politeness_rating/configs/politeness.yaml -p 80
```

Techniques

According to Schulhoff et al. (2024)



tree of techniques

Important Note

There is no substitute for reading Schulhoff et al. (2024)! I'm just listing the main concepts here. I'll ask you to pick one and explain it in your own words.

Top level

- Zero-Shot
- Few-Shot
- Thought Generation
- Ensembling
- Self-Criticism
- Decomposition

Few-Shot Design Decisions

- Exemplar Quantity: as many as possible
- Exemplar Ordering: randomly order them
- Exemplar Label Distribution: balance the distribution
- Exemplar Label Quality: ensure correct labeling
- Exemplar Format: use a common format
- Exemplar Similarity: select similar examples to the test instance

Few-Shot Techniques

- *difficult to implement*
- K-Nearest Neighbors
- Vote-K
- Self-Generated In-Context Learning
- Prompt Mining
- Complicated Techniques use iterative filtering, embedding and retrieval, and reinforcement learning

Zero-Shot Techniques

- *use no exemplars*
- Role Prompting
- Style Prompting
- Emotion Prompting
- System 2 Attention
- SimToM
- Rephrase and Respond
- Re-reading
- Self-Ask

Thought Generation

- *prompting the model to articulate its ongoing reasoning*
- Chain-of-Thought
- Zero-Shot Chain-of-Thought
- Step-Back Prompting
- Analogical Prompting
- Thread-of-Thought Prompting
- Tabular Chain-of-Thought

Few-Shot CoT

- *multiple examples, including chains-of-thought*
- Contrastive CoT Prompting
- Uncertainty-Routed CoT Prompting
- Complexity-based Prompting
- Active Prompting
- Memory-of-Thought Prompting
- Automatic Chain-of-Thought Prompting

Decomposition

- *explicitly decomposing the problem into subproblems*
- Least-to-Most Prompting
- Plan-and-Solve Prompting
- Tree-of-Thought Prompting
- Recursion-of-Thought Prompting
- Program-of-Thoughts
- Faithful Chain-of-Thought
- Skeleton-of-Thought
- Metacognitive Prompting

Ensembling

- *using multiple prompts to solve the same problem, then aggregating the results, for example, by majority vote*
- Demonstration Ensembling
- Mixture of Reasoning Experts
- Max Mutual Information Method
- Self-Consistency
- Universal Self-Consistency
- Meta-Reasoning over Multiple CoTs
- DiVeRSe
- Consistency-based Self-Adaptive Prompting
- Universal Self-Adaptive Prompting
- Prompt Paraphrasing

Self-Criticism

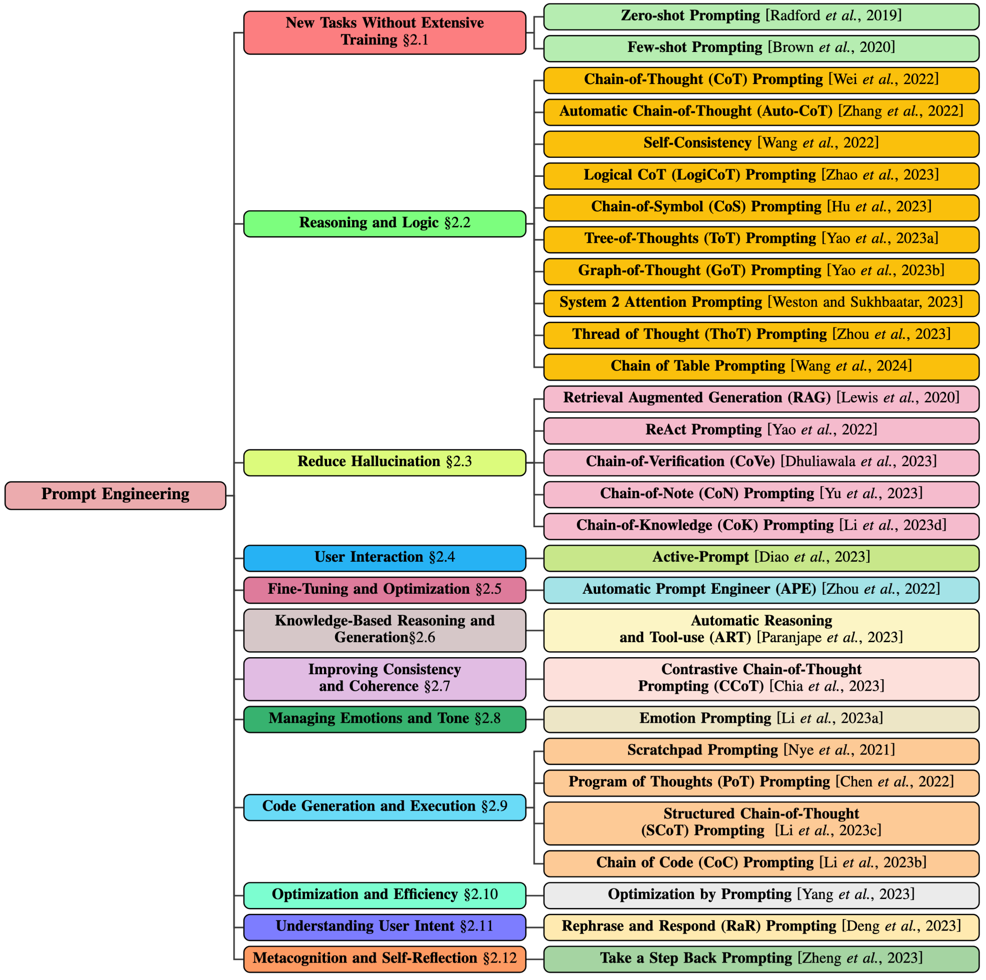
- *prompting the model to critique its own output*
- Self-Calibration
- Self-Refine
- Reversing Chain-of-Thought
- Self-Verification
- Chain-of-Verification
- Cumulative Reasoning

Prompt Engineering Techniques and Goals

Prompt Engineering Techniques

- Meta Prompting
- AutoPrompt
- Automatic Prompt Engineer
- Gradientfree Instructional Prompt Search
- Prompt Optimization with Textual Gradients
- RLPrompt
- Dialogue-comprised Policy-gradient-based Discreet Prompt Optimization

Tree of Prompt Engineering Techniques from Sahoo et al. (2024)



tree of prompt engineering techniques

Goals of Prompt Engineering, Sahoo et al. (2024), 1 of 2

- New tasks without extensive training
- Reasoning and logic
- Reduce hallucination
- User interaction
- Fine-tuning and optimization
- Knowledge-based reasoning and generation
- Improving consistency and coherence

Goals of Prompt Engineering, Sahoo et al. (2024), 2 of 2

- Managing emotions and tone
- Code generation and execution
- Optimization and Efficiency
- Understanding User Intent
- Metacognition and Self-Reflection

Automatic Prompt Optimization, Wan et al. (2024)

- Prompt engineering can be automated through APO
- Two approaches:
- → Instruction Optimization
  → Exemplar Optimization
- They are synergistic, but
- → EO is easier and more effective

Frameworks

Haystack

⟨ work through the tutorial ⟩

END

References

Sahoo, Pranab, Ayush Kumar Singh, Sriparna Saha, Vinija Jain, Samrat Mondal, and Aman Chadha. 2024. "A Systematic Survey of Prompt Engineering in Large Language Models: Techniques and Applications." https://arxiv.org/abs/2402.07927.

Schulhoff, Sander, Michael Ilie, Nishant Balepur, Konstantine Kahadze, Amanda Liu, Chenglei Si, Yinheng Li, et al. 2024. "The Prompt Report: A Systematic Survey of Prompting Techniques." https://arxiv.org/abs/2406.06608.

Wan, Xingchen, Ruoxi Sun, Hootan Nakhost, and Sercan O. Arik. 2024. "Teach Better or Show Smarter? On Instructions and Exemplars in Automatic Prompt Optimization." https://arxiv.org/abs/2406.15708.

Colophon

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